

DSA I – Solving Session

Time Complexity Analysis

by Mahfuz Hasan Reza

b) Discuss the time complexity of the following algorithm.

[3]

```
sum=0;
for(i=1; i<=n; i++){
    for(j=1; j<=n; j++){
        sum=sum+i+j;
    }
}
printf("%d", sum);
```

DSA I - Solving Session Time Complexity Analysis

Previous Question Solve
by Mahfuz Hasan Reza

* Mid Exam, Year: 2023, Trimester: Spring

①(b)

```
sum=0;  $\longrightarrow 1$ 
for(i=1; i<=n; i++){  $\longrightarrow (n+1)$ 
    for(j=1; j<=n; j++){  $\longrightarrow n \times (n+1)$ 
        sum=sum+i+j;  $\longrightarrow n \times (n)$ 
    }
}
printf("%d", sum);  $\longrightarrow 1$ 
```

$$\begin{aligned} \text{Time function, } T(n) &= 1 + n + 1 + \tilde{n} + n + \tilde{n} + 1 \\ &= 2\tilde{n} + 2n + 3 \end{aligned}$$

$$\therefore \text{Time complexity} = O(n^2)$$

b) Discuss the time complexity of the following algorithm.

```
sum=0;
for(i=1; i<=n; i++){
    for(j=1; j<=i; j++){
        sum=sum+i+j;
    }
}
printf("%d", sum);
```

* Mid Exam, Year : 2022, Trimester: Fall

(1)(b)

sum=0; $\xrightarrow{1}$
 for(i=1; i<=n; i++){ $\xrightarrow{(n+1)}$
 for(j=1; j<=i; j++){ $\xrightarrow{\frac{(n+1)(n+2)}{2} - 1}$
 sum = sum + i + j; $\xrightarrow{\frac{n(n+1)}{2}}$
 }

printf("%d", sum); $\xrightarrow{1}$

For inner loop:

i :	1	2	3	4	5	...	n	(n+1)
head :	2	3	4	5	6	...	(n+1)	
body :	1	2	3	4	5	...	n	

$$\therefore \text{head} : 1 + 2 + 3 + 4 + 5 + 6 + \dots + (n+1) - 1$$

$$= \frac{(n+1)(n+2)}{2} - 1$$

$$\therefore \text{body} : 1 + 2 + 3 + 4 + 5 + \dots + n$$

$$= \frac{n(n+1)}{2}$$

$$\therefore \text{Time Function, } T(n) = 1 + (n+1) + \frac{(n+1)(n+2)}{2} - 1 + \frac{n(n+1)}{2} + 1$$

$$\therefore \text{Time Complexity} = O(n^2)$$

b) Discuss the time complexity of the following algorithm.

[2]

```
sum=0;
```

```
for(i=1; i<=n; i++){
```

```
    for(j=1; j<=i; j++){
```

```
        sum=sum+i+j;
```

```
    }
```

```
}
```

```
printf("%d", sum);
```

(n²)

* Mid Exam, Year: 2022, Trimester: Spring

(1) (b)

Same as Fall-2022 (Mid Exam) - 1(b)

b) Discuss the time complexity of the following algorithm.

```

sum=0;
for(i=0; i<n; i++){
    scanf("%d", &A[i]);
    j=n-1;
    while(j>=0){
        sum=sum+A[i]+A[j];
        j--;
    }
}

```

* Mid Exam, Year : 2021, Semester : Summer
1(6)

$sum=0;$ $\longrightarrow 1$
 $for(i=0; i<n; i++){$ $\longrightarrow (n+1)$
 $scanf("%d", \&A[i]);$ $\longrightarrow n$
 $j=n-1;$ $\longrightarrow n$ ~~n~~
 $while(j \geq 0){$ $\longrightarrow n \times (n+1)$
 $sum=sum+A[i]+A[j];$ $\longrightarrow n \times n$
 $j--;$ $\longrightarrow n \times n$
 $}$
 $}$

$\therefore$ Time function, $T(n) = 1 + (n+1) + n + n + n(n+1) + n^2 + n^2$

$$= 3n^2 + 4n + 2$$

$\therefore$ Time Complexity, $O(n^2)$

4. How many times each of statements and conditions will be executed?

```
k=0;
for( i=0; i<=3; i++){
    sum=0;
    for (j=1; j<4; j++){
        sum=sum+i+j;
        k=k+1;
    }
    k=1;
}
```

#Random Problem: (Solve)

⊗ How many times?

$k=0;$ $\longrightarrow$ 1 times
 $\text{for}(i=0; i \leq 3; i++)\{$ $\longrightarrow$ 5 times
 $\text{sum}=0;$ $\longrightarrow$ 4 times
 $\text{for}(j=1; j < 4; j++)\{$ $\longrightarrow$ (4×4) times
 $\text{sum}=\text{sum}+i+j;$ $\longrightarrow$ (4×3) times
 $k=k+1;$ $\longrightarrow$ (4×3) times
 $\}$
 $k=1;$ $\longrightarrow$ 4 times
 $\}$

Total Times: $1 + 5 + 4 + 16 + 12 + 12 + 4$
 $= 54$ times


```

sum=0;
k=10;
for(i=1; i<=n; ++i){
    for(j=3; j<=n; ++j){
        sum=sum+j;
        k=k-1;
    }
    printf("%d", sum);
}
printf("%d %d", sum, k);

```

Random Problem: (Solve)

⊗ Time complexity

$sum = 0;$ $\longrightarrow 1$
 $k = 10;$ $\longrightarrow 1$
 $for(i=1; i \leq n; ++i) \{ \longrightarrow (n+1)$
 $for(j=3; j \leq n; ++j) \{ \longrightarrow n \times (n-1)$
 $sum = sum + j; \longrightarrow n \times (n-2)$
 $k = k - 1; \longrightarrow n \times (n-2)$
 }
 $printf("%d", sum); \longrightarrow n$
 $\}$
 $printf("%d %d", sum, k); \longrightarrow 1$

$$\begin{aligned}
 \text{Time function, } T(n) &= 1 + 1 + (n+1) + (n^2 - n) + (n^2 - 2n) + \\
 &\quad (n^2 - 2n) + n + 1 \\
 &= 3n^2 - 3n + 4
 \end{aligned}$$

$\therefore$ Time Complexity, $O(n^2)$


```

sum=0;
k=0;
while(k<=n){
    sum=sum+k;
    printf("%d %d", sum, k);
    k=k+1;
}
printf("%d", sum);

```

Random Problem: (Solve)

⊛ Time Complexity

$sum=0;$ $\longrightarrow 1$
 $k=0;$ $\longrightarrow 1$
 $while(k \leq n)\{$ $\longrightarrow (n+2)$
 $sum=sum+k;$ $\longrightarrow (n+1)$
 $printf("%d %d", sum, k);$ $\longrightarrow (n+1)$
 $k=k+1;$ $\longrightarrow (n+1)$
 $\}$
 $printf("%d", sum);$ $\longrightarrow 1$

Time function, $T(n) = 1 + 1 + (n+2) + (n+1) + (n+1) + (n+1) + 1$

Time function, $T(n) = 4n + 8$

$\therefore$ Time Complexity, $O(n)$